

More public measurements for rigid-body contact

Fixed material inputs, new comparisons and prediction readiness

October 7, 2026

The project now has additional rubber-ball, sliding, non-spherical planar impact, spatial impact and limestone rocking observations. This checkpoint holds published material metadata fixed and exposes mismatches. It does not establish a complete model that reproduces real collisions. No material coefficients were fitted here.

Actual comparisons. Errors below are predicted minus measured. Trial 1 is an exploratory pilot in each GAUGE family/material and is excluded from evaluation. Multiple samples/events within a trial are correlated. RMSE values for different observables have different units and must not be pooled.

Comparison	Evaluation amount	RMSE / maximum error
Rubber bounce; observed impact marker	42 events in 21 trials	0.380733 / 0.671462 m/s
Wood sliding; nominal slope	19 trials, 167 samples	6.962 / 15.424 mm
Plastic sliding; nominal slope	19 trials, 153 samples	10.824 / 31.245 mm
Metal sliding; nominal slope	18 trials, 158 samples	2.880 / 7.771 mm
Limestone; ideal rocking geometry	134 unique first events	0.039013 / 0.125871 ratio

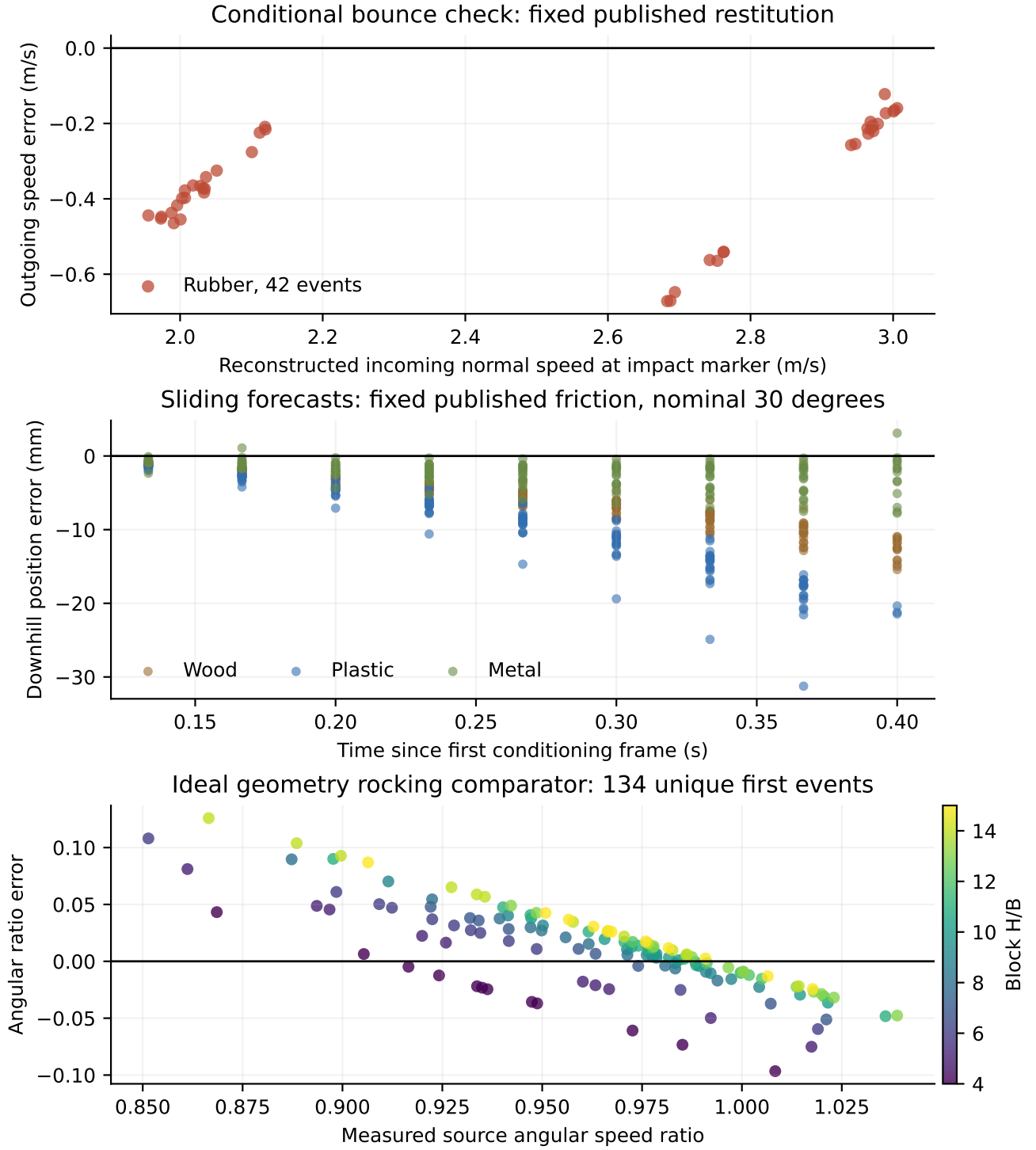
Sliding forecasts condition position and velocity on four initial moving samples, then predict subsequent positions using fixed friction and gravity. The support window ends before 0.30 m displacement from the initial sample. There are 59 released trials and 56 evaluation trials. The material values are unchanged; their source field does not separately identify static and dynamic friction.

The bounce check conditions on an observed impact-marker timestamp, infers incoming velocity from the incoming flight only, and compares the predicted outgoing speed with an outgoing-only flight reconstruction at that timestamp. It does not predict the contact time or use independently measured high-rate contact states. The source restitution underpredicts the reconstructed response: mean signed error -0.350346 m/s.

State imports, not additional endpoint predictions. MIT supplies 1,718 signed planar before/after states and contact Jacobians. GAUGE supplies 160 spatial tetrahedron, wedge and pyramid trials: 280 body records and 4,686 pose samples. These are now imported with explicit flat indices and units. Missing matched coefficients, spatial inertia and contact-state uncertainty prevent a claim of complete fixed-input prediction for these imports.

Prediction status. This is a normal-branch check, sustained-sliding forecast, ideal-rocking comparator and data extension. The full directional impulse/friction law and production solver have not been validated on these new data. The attempted measured-angle correction worsens every sliding trial.

Deviations from measured reality



The zero line means agreement. Colors distinguish materials or block geometry; every admitted evaluation point is shown. Separate panels retain physical units. Bounce and sliding are branch comparisons; the rocking panel is an established ideal-geometry comparator. No bar charts or outcome-based removal of outliers. The bounce timing envelope is discussed on the last page and is not drawn as a statistical uncertainty band.

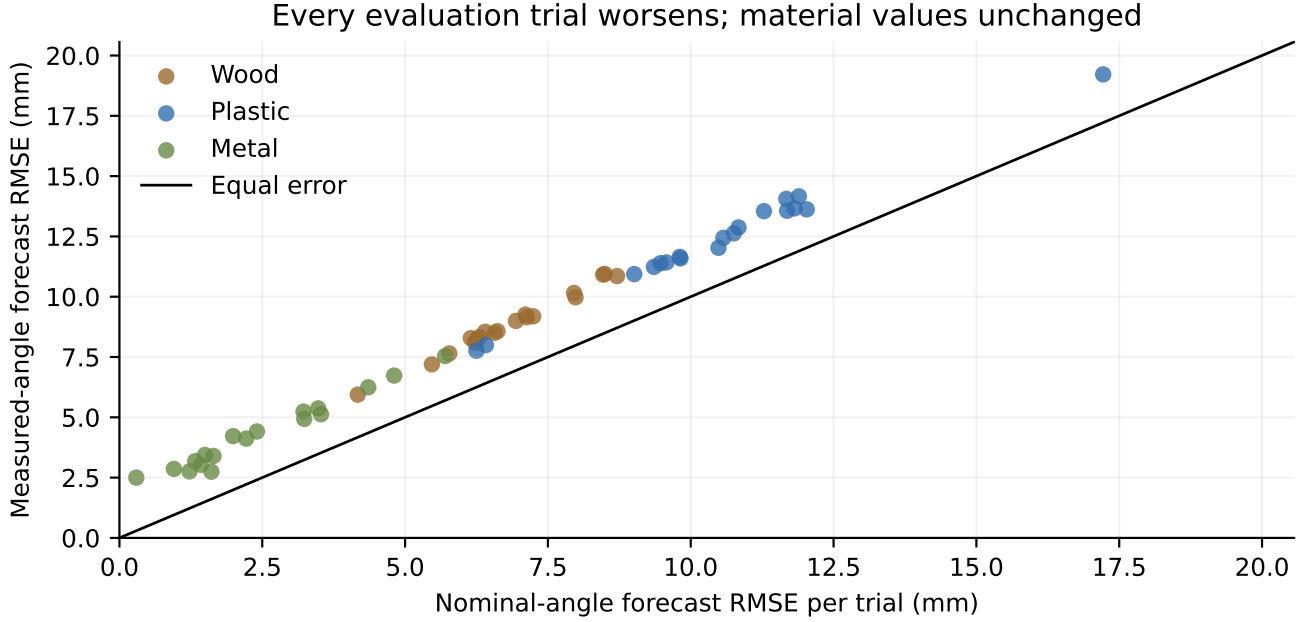
Published inputs and the failed correction

Case	Values held fixed	Provenance / missing information
GAUGE rubber	Mass 0.121 kg; source restitution 0.576733	Generic restitution field; calibration-pair and split details incomplete
GAUGE sliders	Friction: wood 0.273673; plastic 0.275018; metal 0.262758	Moving-branch interpretation; no separate μ_s, μ_d
MIT ellipse	Mass 0.0364 kg; semiaxes 0.035/0.025 m; gyration radius 0.0192 m	Reported planar $\mathbb{I} = 1.3418496 \cdot 10^{-5} \text{ kg m}^2$; independent matched contact coefficients absent
Limestone blocks	Nominal dimensions, mass, geometric inertia	Effective geometry/ratios are outcomes; energy correction $f_c = 0.7$ is assumed

In symbols, the two checked branches use

$$v_n^+ = e_{\text{source}} |v_n^-|, \quad a = g \sin \theta - \mu g \cos \theta, \quad x = x_0 + v_0 t + \frac{1}{2} a t^2.$$

Numerical material values are separate from the equations. Published scalar fields do not automatically supply all normal/tangential restitution, static, dynamic, rolling or spin resistance parameters of the full model.



The candidate uses board quaternions from the same four conditioning frames, projects gravity onto the measured plane and leaves friction unchanged. Quaternion and source Euler orientations agree within $1.74 \cdot 10^{-6}$ rad. Measured slopes are 29.56–29.67 degrees. Candidate aggregate RMSE becomes 9.010/12.709/4.573 mm (wood/plastic/metal), with **0/56 improved trials**. This correction cannot explain the discrepancy. Retain the measured geometry and investigate contact or calibration mismatch; do not change the angle or friction to hide the errors.

Non-spherical, spin and deformation challenges

Planar ellipse impacts. The signed pre/post translation and angular velocity are directly useful for testing coupled momentum transfer. The source reports 250 Hz sampling and an experiment confined by a glass guide. Its supplied contact Jacobians permit the diagnostic reconstruction

$$\Delta \mathbf{p} = m \Delta \mathbf{v}, \quad \delta \mathbf{L}_{\text{residual}} = \Delta \mathbf{L} - \mathbf{r} \wedge \delta \mathbf{p}.$$

The residual couple RMS is 0.00118023 N m s, but this is inferred from outcomes, not independently measured contact torque. Finite-contact timing, contact-point changes, gravity/external impulses, guide reactions and measurement noise can contribute. Nominal ellipse geometry differs from the supplied contact Jacobians by up to 1.72 mm. Do not identify the residual with a measured material coefficient. The source authors fitted their model coefficients on these same data; those are not independent inputs for our prediction test.

Spatial tetrahedron, wedge and pyramid impacts. All 160 released trials are imported, not just schema pilots. Translation is converted from mm to m; source quaternions remain in xyzw order, with norm error below $9 \cdot 10^{-7}$. No measured full inertia tensor is supplied. Mesh-based homogeneous inertia would be an explicit assumption. Source folder labels task-1/2/3 differ from metadata task-3/4/5; restitution dictionaries are retained without an inferred mapping. Full signed velocities and short contact intervals need uncertainty-aware reconstruction before spatial endpoint testing.

Limestone rocking. Colombo et al.'s 2026 release contains 135 trial folders, whereas its associated paper describes 120 tests. One processed pair, BlockGroup_1_n13, exactly duplicates BlockGroup_1_n7. Keep both in the source record but use only 134 unique first finite angular-ratio estimates for the primary comparison. The ideal planar pivot-transfer comparator is

$$r_{\omega} = 1 - \frac{3B^2}{2(B^2 + H^2)}.$$

It requires ideal geometry and no sliding/bouncing. Ratios above one are retained: 22/134 primary outcomes and 2,820/7,519 finite unique-cycle estimates. These processed ratios are not certified material restitution. Late-cycle estimates are correlated and can be sensitive to small motions/processing. The archive also contains raw 3D time histories; only processed summaries were used here.

Compliant silicone and an ABS shell. Rémond et al. measure outgoing spin for a 40 mm, 2.7 g shell on glass with silicone layers 0.50, 0.96, 1.25 and 1.78 mm thick. This is a useful challenge for small local shear/pressure history instead of deforming every body. However, their friction near 0.92, effective mass and local stiffness are fitted from rebound/spin results; derived moduli are not independent characterization. Figure 8 has not yet been digitized here, so no new prediction or numeric data count is claimed for this experiment.

The user model retains independent $\delta \mathbf{L}$, not just a lever moment, and the full relative contact-velocity direction $\hat{\mathbf{t}}$ and spin direction $\hat{\mathbf{s}}$. Conventional comparators do not resolve the zero-slip directional closure or certify the complete implementation.

Measurement audit, reproducibility and next use

Bounce input dependency corrected. The original flight-intersection reconstruction used both incoming and outgoing arcs to infer contact time. Consequently the outgoing trajectory indirectly affected its reconstructed incoming state. Its RMSE 0.382254 m/s is retained as a joint-arc consistency diagnostic, not an independent endpoint prediction. The main table instead uses the observed valley timestamp and incoming-only coefficients. This correction was made after inspecting the initial outcomes; no pristine blind validation is claimed. Restitution never changed.

Released GAUGE JSON is 30 Hz although the paper describes 180 Hz capture. A conservative one-frame marker-time sensitivity for the signed speed error is $\pm g(1 + e_{\text{source}})/30 = \pm 0.515592$ m/s. It spans zero for 35/42 corrected errors. This envelope is not a confidence interval. Flight fits can be very good without identifying true compression duration or deformable surface contact height. Distinguish source-parameter mismatch from unresolved impact state and boundary conditions before adopting a new friction/deformation law.

Controls and preserved evidence. Twenty-four analytic, instantaneous bounce controls recover velocities within $1.47 \cdot 10^{-14}$ m/s; ballistic fits within $1.34 \cdot 10^{-15}$. Nine plane/gravity controls pass. These validate the extractor under its ideal assumptions. The MIT case/body-index naming correction is independently checked across all 1,718 rows; physics, all other CSVs and metrics are unchanged. Original evidence, protocols, hashes and source snapshots remain.

Efficient representation. The spatial import uses a flat case index, one body index per case and explicit pose offsets/counts. Separate observed position/quaternion channels can later feed Vektor Flow kernels without dense pair-body tensors. This is data preparation, not a port, throughput benchmark or group simulation. Earlier local contact-kernel benchmarks remain separate.

Next experiment. Resolve the GAUGE scene/material/subtask mapping; reconstruct impact states with sampling and pose uncertainty; then compare the current model with a passive, small local shear-history extension on declared trial splits. Keep documented coefficients fixed. Missing-property estimation, if needed, must have physical bounds and separate evaluation. Do not add an arbitrary impulse polynomial or treat same-outcome fitted coefficients as public independent material measurements. No empirical accuracy gain is claimed here.

Primary sources.

- Wang et al., [GAUGE \(2026\)](#); [dataset](#), MIT license.
Pinned revision 9e0acb70fecc0d4161660264d9a4b08d8f56d45a.
- Fazeli et al., [Learning Data-Efficient Rigid-Body Contact Models \(2017\)](#); [author dataset](#).
Pinned revision f24a7e3b31ad0b53652d6b2a6b26a702cc4362da. Original MAT/code not redistributed here; derived observations attributed.
- Colombo et al., [Free-rocking masonry dataset](#), DOI 10.60756/uminho-jh25, CC BY 4.0; [associated paper](#). Archive SHA-256 a2423df131acf78692a18f8732d4054b1e87b5d798f8d322adb5d96d773952f4.
- Rémond et al., [Effect of a compliant substrate on the rebound of a spherical shell \(2026\)](#), DOI 10.1103/mmdr-2mm3; original PDF outside package.

Download receipts, per-row CSVs, flat pose arrays, scripts, protocols, source catalog and this PDF accompany the report package. See the repository README for reproduction commands. Original large archives remain in an external cache. Scope: research-data extension; production physics unchanged.